

Olist E-Commerce Logistics & Customer Experience Analysis

Business Insights Report

Project Overview

This project looks at the Olist e-commerce dataset with a focus on customer experience and logistics performance. I wanted to understand what factors are connected to customer dissatisfaction and whether issues such as delivery delays, freight costs, seller performance, and product categories have an impact on customer reviews.

For the analysis, I used SQL, Python, and Power BI. The final dashboard has two pages. The first page gives a general overview of orders, delivery performance, and customer satisfaction. The second page allows a deeper look into seller risk, freight costs, delivery delays, and customer review patterns.

Overall, the dataset contains around 96K orders. The on-time delivery rate is 91.9%, the average customer review is 4.16, and 12.8% of customers are dissatisfied. At first glance, these numbers show that the overall performance is quite good, but there are still some areas where the business could improve.

Key Business Insights

1. Delivery delays have a clear effect on customer reviews

One of the strongest patterns in the analysis is the relationship between delivery performance and customer satisfaction. Orders delivered on time have the highest average review scores. As delivery delays increase, the average review score goes down. High-delay orders have the poorest customer experience. This suggests that delivery performance is not just an operational issue, it directly affects how customers rate their overall experience.

2. Most orders are delivered on time, but the delayed orders still create a customer experience problem This suggests that delivery performance is not just an operational issue; it directly affects how customers rate their overall experience.

The overall on-time delivery rate is **91.9%**, which is a strong result. However, the remaining delayed orders are important because they have much lower review scores. Even though delayed orders represent a smaller part of the total volume, they are more likely to create dissatisfied customers. This means improving the experience of this smaller group could have a noticeable impact on the overall customer satisfaction rate.

3. Freight costs are higher for dissatisfied customers

The analysis shows that dissatisfied customers generally paid higher freight costs compared with satisfied customers. The freight-to-product-price ratio is also slightly higher for dissatisfied customers. This is important because customers may be less accepting of a poor experience when they feel that shipping costs are too high compared with the value of the product.

4. Seller performance is not the same across all risk segments

The seller segmentation analysis shows clear differences between seller groups. Strong performers handle the largest order volume and maintain good customer reviews. Other segments, such as **Critical Seller**, **Customer Product Risk**, and **Delivery Risk**, show different types of potential problems. This means the business should not treat all sellers in the same way. Different risk groups may require different actions.

5. Customer and product risk cannot be explained only by delivery

The Customer Product Risk segment shows weaker customer experience compared with stronger seller groups. This suggests that not every dissatisfaction issue is caused by late delivery. Some problems may come from the product itself, customer expectations, seller service, or a combination of these factors. Looking only at delivery performance would therefore miss part of the problem.

6. Delivery risk needs separate attention

The Delivery Risk segment also shows a weaker customer experience. This points more directly towards logistics-related issues. For these orders, the problem may involve shipping delays, delivery partners, or the gap between the expected and actual delivery experience. Separating this segment from other seller risks helps identify where the problem is more likely to come from.

7. Some product categories have noticeably higher dissatisfaction rates

The dashboard highlights several product categories with higher dissatisfaction rates than others. Categories such as **movies and audio** appear near the top of the list. These categories should be investigated further. The higher dissatisfaction could be related to product quality, customer expectations, seller performance, delivery issues, or other category-specific factors. Instead of trying to improve every category equally, the business could focus first on the categories showing the highest level of customer dissatisfaction.

8. Customer satisfaction changes over time

The monthly trend shows that customer satisfaction is not completely stable. There are periods where the dissatisfaction rate increases and average review scores drop. This shows why looking only at the overall average can sometimes hide problems. Regular monitoring would make it easier to identify unusual changes and investigate what caused them.

Summary

The overall results are positive, especially the **91.9% on-time delivery rate** and **4.16 average review score**. However, the analysis also shows clear areas of risk. The main pattern is that **late delivery is strongly connected with lower customer satisfaction**. Higher freight costs, certain seller risk groups, and some product categories also appear to contribute to dissatisfaction. The findings suggest that the business should focus on the specific areas creating problems instead of applying the same solution everywhere.

Business Recommendations

1. Focus first on reducing high-delay orders

Since high-delay orders have the lowest customer review scores, reducing severe delivery delays should be a priority. The business should identify where these delays are happening, such as specific sellers, locations, or logistics partners. Orders that are already at risk of being delayed could also be monitored earlier so that action can be taken before the customer experience becomes worse.

2. Give customers better communication when delays happen

Not every delivery delay can be avoided, but poor communication can make the experience even worse. Customers should receive clear updates when an order is delayed. Providing a revised delivery estimate can help manage expectations and reduce frustration.

3. Review freight costs for low-value products

The analysis shows that dissatisfied customers tend to face higher freight costs and a higher freight-to-product-price ratio.

The business should review cases where shipping costs represent a large part of the product price. Possible solutions could include seller shipping incentives, better logistics options, or promotional shipping offers for selected products.

The goal should not simply be to reduce all freight costs, but to identify situations where customers are paying a relatively high amount for delivery compared with the value of the product.

4. Manage seller segments differently

The seller segmentation shows that different groups have different risk profiles. Because of this, a single approach for all sellers would not be effective.

- **Strong performers** should be monitored and used as examples of good performance.

- **Critical sellers** should receive closer monitoring and clear improvement targets.
- **Customer product risk sellers** should be investigated for possible product or customer experience issues.
- **Delivery risk sellers** should be reviewed together with logistics performance data.

This would allow the business to focus its attention where it is most needed.

5. Investigate high-risk product categories

Product categories with the highest dissatisfaction rates should be reviewed in more detail. The business should check whether the problem is connected to product quality, inaccurate product descriptions, seller performance, delivery issues, or customer expectations. A useful next step would be to create a category-level monitoring process for the highest-risk categories instead of treating all categories equally.

6. Create an early warning system for customer dissatisfaction

The dashboard already identifies several useful warning signals, including delivery delays, freight costs, seller risk, and product category dissatisfaction. These metrics could be combined into a simple monitoring framework. For example, orders could be flagged when they have a high risk of delay, unusually high freight cost, or come from a higher-risk seller segment. This would help the business move from only analyzing problems after they happen to identifying potential issues earlier.

7. Track customer satisfaction regularly instead of relying only on averages

The monthly trend shows that customer satisfaction can change over time. The business should monitor metrics such as average review score, dissatisfaction rate, on-time delivery rate, and seller risk on a regular basis. Sudden changes should be investigated instead of waiting for them to affect the overall yearly performance.

Recommendation Priority

If the business had to focus on only three areas first, I would prioritize:

1. **Reducing severe delivery delays**
2. **Improving high-risk seller and product category performance**
3. **Reviewing cases where freight costs are high compared with product value**

These three areas are the most directly connected to the customer dissatisfaction patterns found in the analysis.

Tools & Technologies Used

SQL

Used for data exploration, joining tables, and answering business questions from the Olist dataset.

Python

Used for data cleaning, transformation, exploratory analysis, and creating calculated metrics such as delivery delays, customer dissatisfaction, freight ratios, and risk segments.

Main libraries used:

- Pandas
- NumPy
- Matplotlib / Seaborn

Power BI

Used to build the final interactive dashboard, create DAX measures, KPIs, slicers, risk analysis, and visualizations.

GitHub

Used to document the project and store the SQL queries, Python analysis, Power BI files, and project documentation.

Conclusion

This project shows how data can be used to move beyond basic reporting and understand the factors behind customer dissatisfaction.

The analysis found that delivery delays are the strongest customer experience risk, while freight costs, seller performance, and product categories also contribute to different levels of customer dissatisfaction.

The final Power BI dashboard brings these findings together in an interactive format, allowing users to move from an overall business view to a more detailed risk analysis.

Overall, this project demonstrates an end-to-end analytics workflow:

SQL → Python → Data Analysis → Power BI → Business Insights → Recommendations